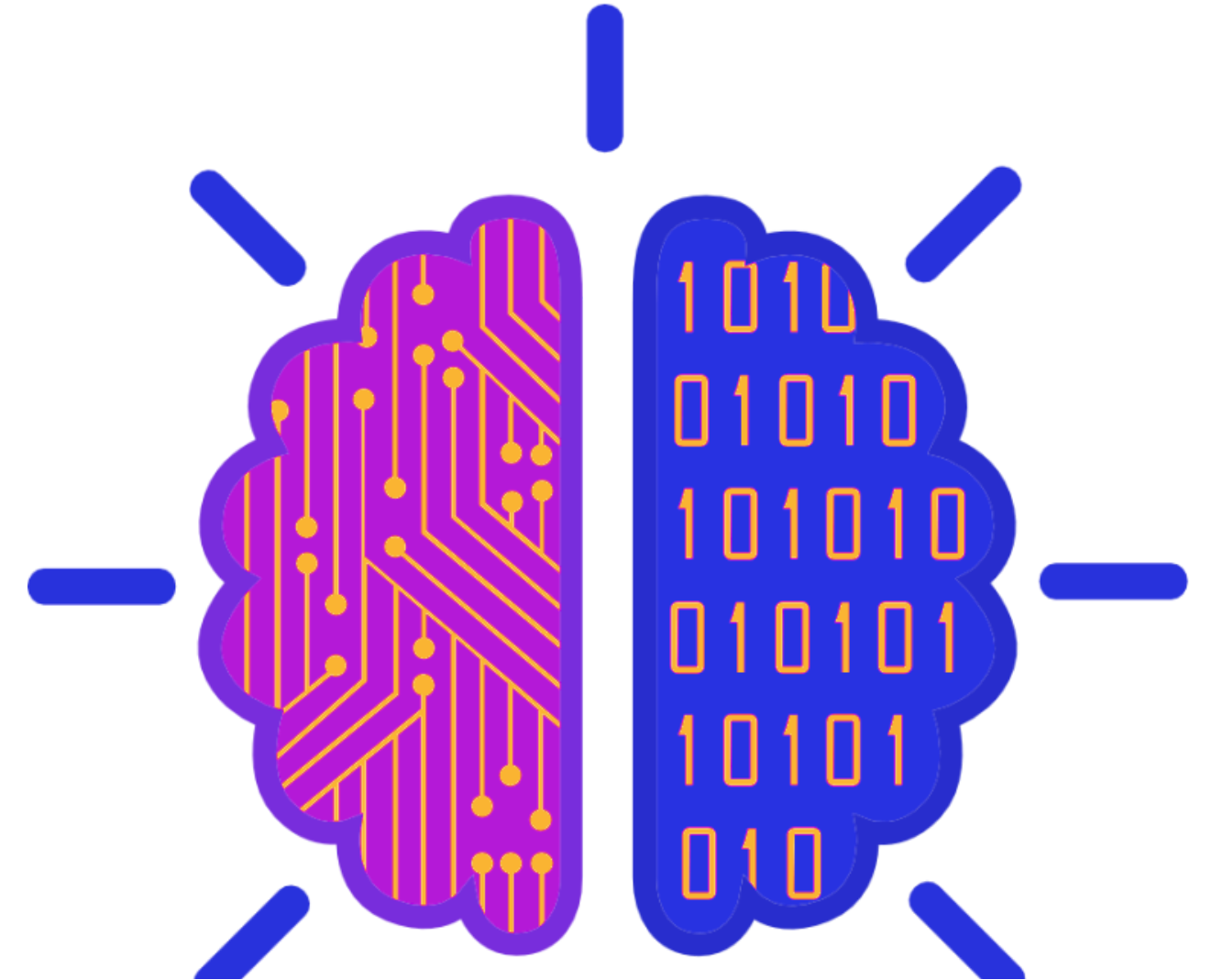


MAKEATHON 3.0

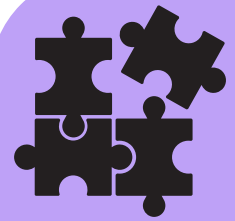
II YEAR

- **Problem Statement ID: SIH25050**
- **Problem Statement Title: Smart Traffic Management System for Urban Congestion**
- **Theme: Transportation & Logistics**
- **PS Category: Software**
- **Team Name: Tech Thunders**



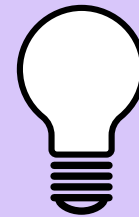
URBAN FLOW ORCHESTRATOR (UFO)

From Reactive Signals to Proactive City Orchestration



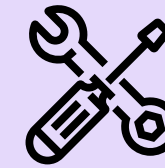
PROBLEM

- **Reactive Signals:**
Lights change only after jam forms
- **Emergency Delays:**
Ambulances stuck, golden minutes lost
- **Data Silos:**
No integration with events, weather, schools
- **Untapped Public Transport:**
Buses wait like private cars
- **Blind Commuters:**
No real-time alternate route guidance



IDEA

- **Predictive AI Engine:**
Forecasts congestion before it forms
- **Proactive Orchestration:**
Shifts from reactive to city-wide control
- **Unified Integration:**
Connects emergency, public transport, and city databases



SOLUTION

- **Predictive Engine:**
Analyzes real-time data + event schedules + weather forecasts
- **Priority Pre-emption:**
Creates green corridors for ambulances and late-running buses
- **V2I Communication:**
Suggests alternate routes to drivers via navigation apps
- **Unified Dashboard:**
Single view for traffic authorities to monitor and control



UVP

- **Predictive, not Reactive:**
Predicts congestion using city event data and weather APIs
- **Life-Saving Priority:**
Reduces emergency response time by 25-30%
- **Public Transport Boost:**
Improves bus reliability, encouraging usage
- **Citizen Engagement:**
Directs drivers to alternate routes via apps
- **Hardware Agnostic:**
Works with existing cameras and signals—no new infrastructure

TECHNICAL APPROACH

WORKFLOW

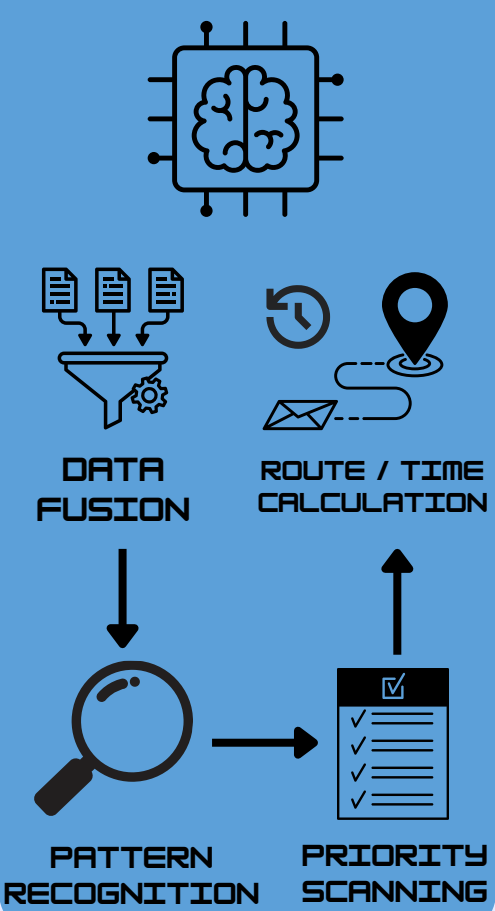
STAGE 1

DATA INGESTION



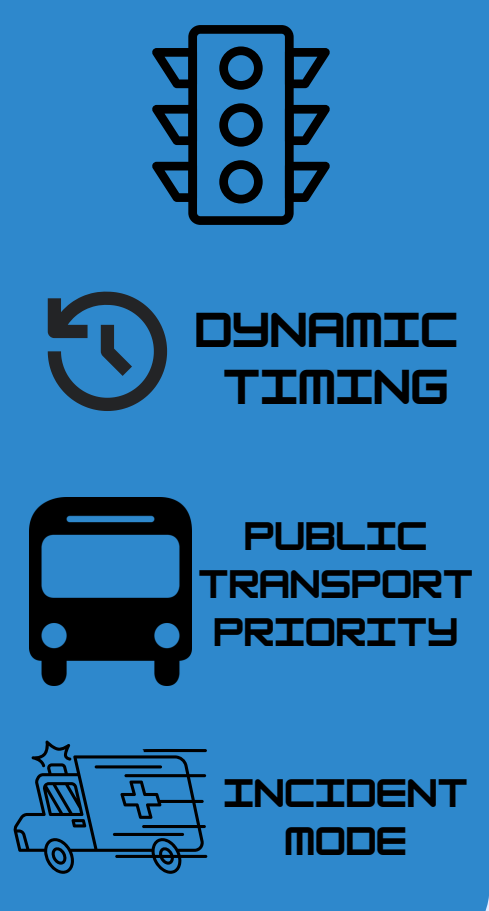
STAGE 2

AI PROCESSING



STAGE 3

SIGNAL CONTROL



STAGE 4

OUTPUT

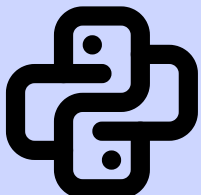


TECH STACK



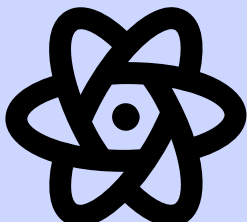
TRAFFIC SIMULATION
SUMO

Simulation of Urban Mobility



PYTHON
FAST API
TraCI

AI BACKEND - Signal Control



NODE.JS
FrontEnd



PostgreSQL
DataBase

FEASIBILITY AND VIABILITY

FEASIBILITY

Hardware Agnostic: Software layer sits on top of existing traffic infrastructure

Modular Integration: APIs easily connect to municipal databases and third-party apps

Proven Algorithms: Predictive models based on existing research (IEEE, Google Traffic)

Scalable Prototype: 7-day prototype using SUMO validates core logic

VIABILITY

Low Capex: No new hardware costs—uses existing cameras and signals

Revenue Model: Annual Maintenance Contracts (AMC) with municipalities

Social ROI: Lives saved (emergency) + man-hours saved (commute) + pollution reduced

Government Alignment: Fits Smart Cities Mission and National Urban Digital Mission

CHALLENGES

Legacy traffic infrastructure

Data privacy concerns

Integration with multiple city depts

Real-time processing latency

Real-time processing latency

SOLUTIONS

Hardware-agnostic design, works with existing systems

Standardized REST APIs for easy adoption

Anonymized vehicle data, no personal tracking

Edge computing at intersections + cloud fallback

Pilot in one corridor to demonstrate 15% travel time reduction

IMPACT AND BENEFITS

IMPACTS

- Lives Saved:** 25-30% faster emergency response through green corridors
- Reduced Pollution:** Less idling = lower fuel consumption and emissions
- Economic Productivity:** Millions of man-hours saved annually in urban commute
- Data-Driven Planning:** Provides city planners with actionable traffic insights

Stakeholder	Benefit
Citizens	Faster commutes, safer roads, reliable public transport
Emergency Services	Guaranteed clear paths during critical moments
Municipalities	Enhanced city image, reduced congestion costs
Environment	Lower vehicular emissions, reduced fuel wastage
Public Transport	Improved punctuality, higher ridership

MARKET STRATEGY

- **Pilot-First Approach:** Deploy in one high-congestion corridor of a mid-tier smart city
- **Performance-Based Scaling:** Demonstrate KPIs (15% travel time reduction) to win city-wide contracts
- **PPP Model:** Municipality provides infrastructure; private firm manages AI software via AMC
- **Partnerships:** Collaborate with 108 Ambulance services and public transport corporations

TARGET USERS

- **Primary:** City Traffic Police, Urban Local Bodies (ULBs)
- **Secondary:** Emergency Services (Ambulance, Fire), Public Transport Corporations
- **Tertiary:** Commuters, Navigation Apps (Google Maps, Apple Maps), Urban Planners

RESEARCH AND REFERENCES

S.N O	SOURCES	LINKS
1	Smart Cities Mission	https://smartcities.gov.in/
2	National Urban Digital Mission (NUDM)	https://mohua.gov.in/
3	Multiple Intelligent Control Strategies for Travel-Time Reduction of Connected Emergency Vehicles – IEEE Transactions	10.1109/TITS.2024.3484771
4	BusTr: Predicting Bus Travel Times from Real-Time Traffic	https://arxiv.org/html/2007.00882
5	The road to seamless urban mobility – McKinsey & Co.	https://upgrade.mckinsey.com.br/featured-insights/future-of-cities

